

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Applied Lab One: Data Visualization

Part III: Visualize Some Data

September 18, 2026

Lab

This is *Part III* of the first Applied Lab. *Part I* and *Part II* were assigned earlier this week.

As before, the lab will rely on the 2019 subset of the [World Development Indicators](#) (WDI) data that we used as our example data in class. Those data have ≈ 40 variables on roughly 215 countries in the world, with values measured in 2019. The description of those data is [here](#), and the data itself is [here](#), in `.csv` format.

Part III is the part of the lab assignment that you are required to submit for credit. Your instructions are as follows:

1. From the WDI data, choose three variables: two *continuous* indicators, and one *discrete* variable. To keep things simple in these instructions, we'll refer to the two continuous variables as X and Y , and the discrete variable as D .
2. Using one of the various online / no-code data visualization apps we reviewed in class, visualize and describe the distributions of X , Y , and D individually. That is, create and describe *univariate* visualizations of each of the three variables you choose. As well as the visualizations, describe the distributions / summaries you observe in a sentence or two. Be sure to include a note about which tool you used to create your visualizations.
3. Using an AI tool, visualize and describe the *joint* (bivariate) distributions of the three variables. That is, create and describe plots of:
 - (a) X and D
 - (b) Y and D
 - (c) X and Y

As you did in (1) above, for each of the three visualizations, write a brief sentence or two describing the visualization you create. In addition, be sure to include information about which AI tool you used to create your three visualizations, and *please include the full text of the prompts used to generate the visualizations* as well.

Resources

We discussed various univariate and bivariate plots in our class sessions. But if you want to dig a bit more and understand things better, here are some reasonable (non-technical) places to start (links in blue):

- [Describing Data, Part One](#) and [Describing Data, Part Two](#), from Penn State's STAT 200 course (ignore all the details about MiniTab).
- [Chapter 5: Presenting Data](#) in the University of Edinburgh's *Statistics for Oncology* course.
- Sections 2.1, 2.2, 2.3, and 9.1 of [Significant Statistics](#), at Virginia Tech.

Note that these might be especially helpful for the written / descriptive parts of the lab assignment.

Details

Your completed lab should include a separate title page with your name, the name of the course, the name of the lab, and the date on which it was completed / submitted. The body of the lab should be *double-spaced*, in a conventionally-sized (11 or 12 point) font. Make sure that each of the six visualizations you create are included, and are sufficiently large and of high enough definition to be readable. Each part of the lab should be as short or long as you deem necessary; a rough target might be a total of 7-8 pages for the entire lab, including prompts. You need not cite any sources or include references, though you are welcome to.

The entire lab exercise is mandatory, and is worth a possible 150 points. Part III of the lab is due on or before before **Friday, September 25, 2026 at 11:59 p.m. ET**. Submit your assignment by *creating / converting your assignment into PDF format*, and emailing it to Dr. Zorn (at zorn@psu.edu). Assignments received after the due date will not be accepted.